

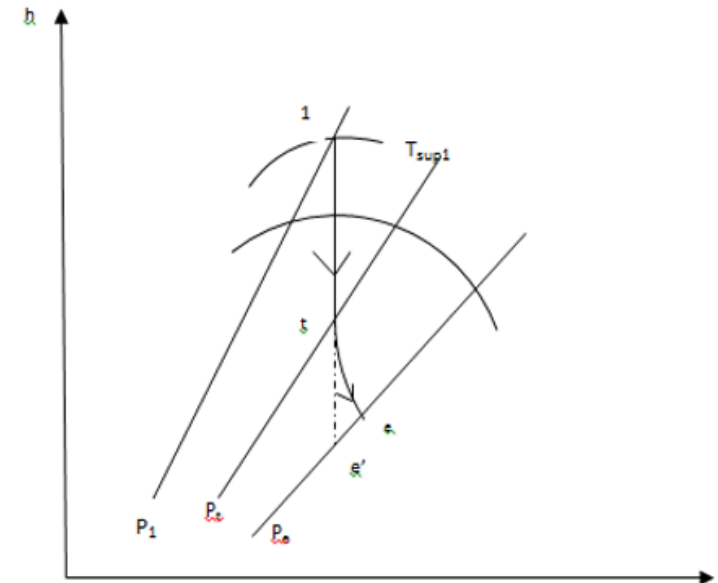


STEAM NOZZLES

Flow of Steam through Nozzle

Effect of Friction on Nozzles:

- Most of the friction occurs in the diverging part of a convergent-divergent nozzle as the length of the converging part is very small.
- It is assumed that the friction losses in convergent portion are negligible.
- The effect of friction is to reduce the available enthalpy drop by about 10 to 15%.



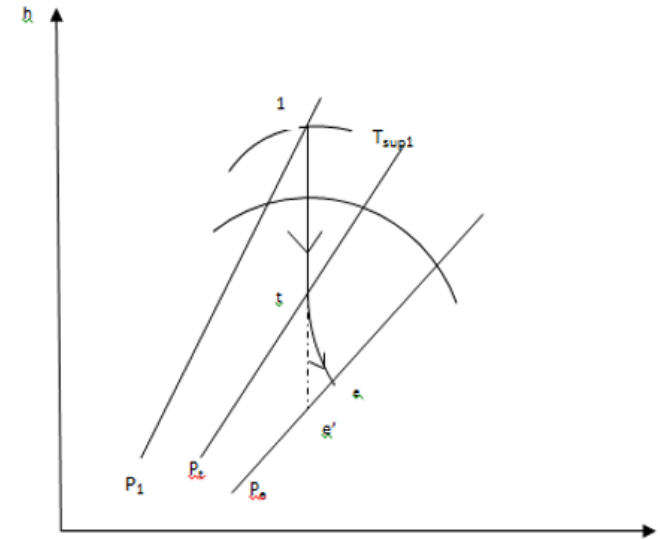
Nozzle efficiency

$$\eta = \frac{h_1 - h_e}{h_1 - h_{e'}}$$

Flow of Steam through Nozzle

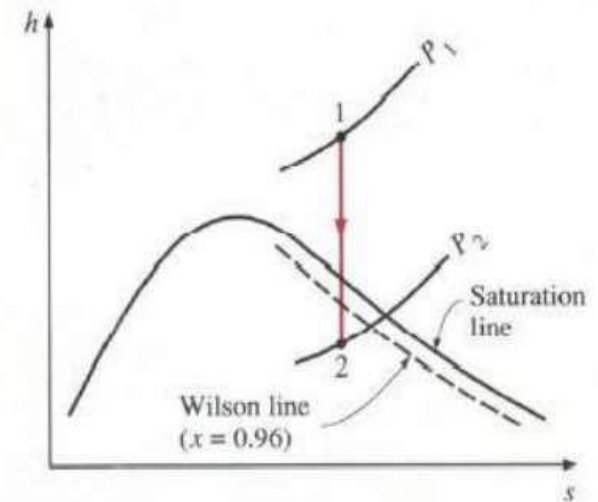
Effect of Friction on Nozzles:

- Entropy is increased.
- The energy available decreases.
- Dryness fraction increases
- Velocity of flow at the exit get decreased (some kinetic energy loss to overcome friction).
- Volume of flowing steam is decreased.



Flow of Steam through Nozzle

- **Super-saturated flow or metastable flow of in Nozzles:**
- As steam expands in the nozzle, the pressure and temperature drops
- Steam starts condensing when it reaches the saturation line.
- But this is not always the situation.



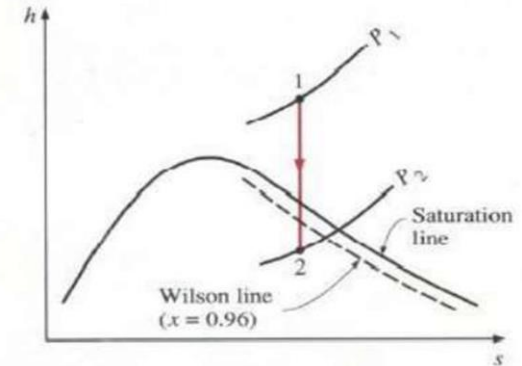
Flow of Steam through Nozzle

- **Super-saturated flow or metastable flow of in Nozzles:**
- Due to the high velocities, the time up to which the steam resides in the nozzle is small
- There may not be sufficient time for the needed heat transfer and the formation of liquid droplets due to condensation.
- As a result, the condensation of steam is delayed for a while.

Flow of Steam through Nozzle

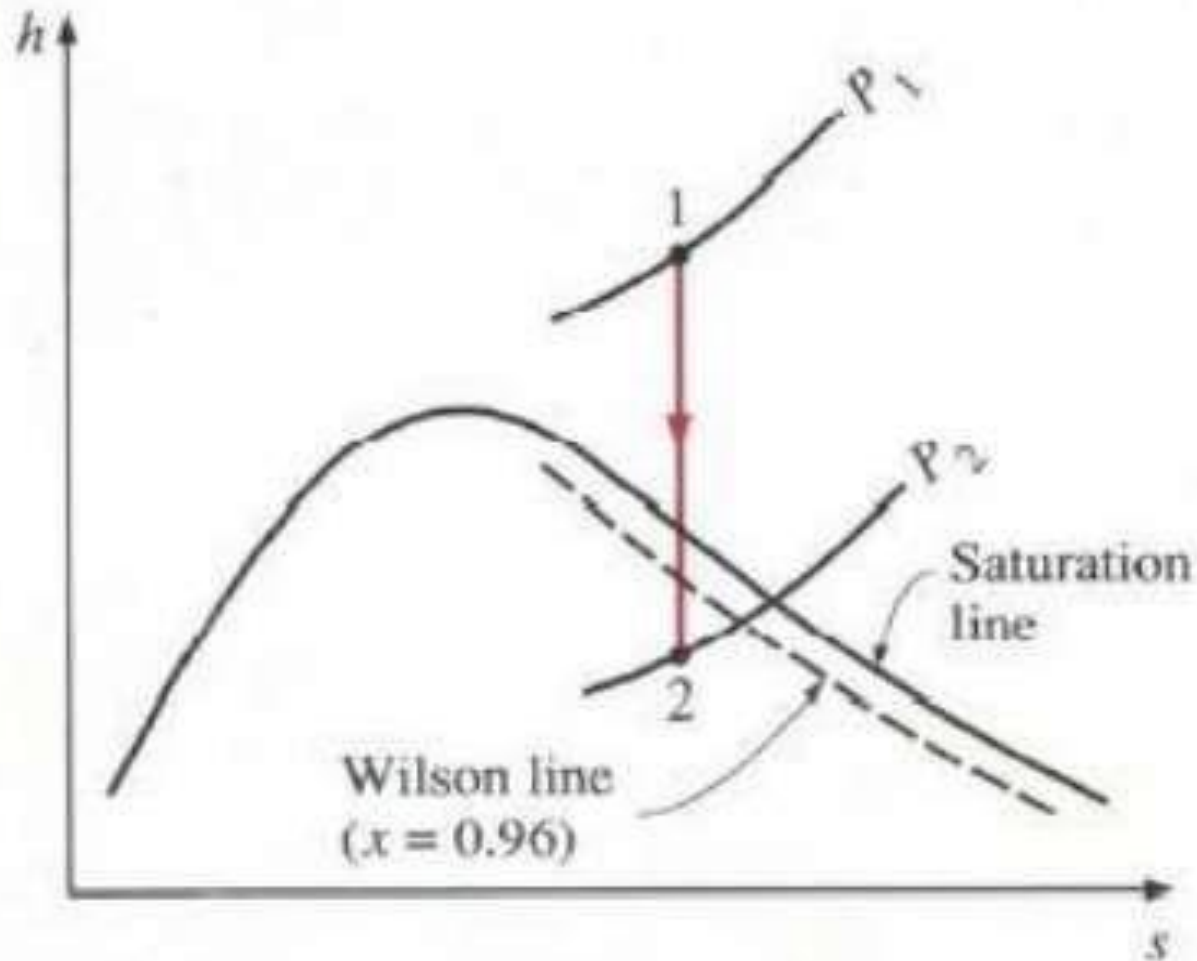
- **Super-saturated flow or metastable flow of in Nozzles:**

- This phenomenon is known as super saturation
- The steam that remains in the wet region without holding any liquid is known as supersaturated steam.
- The locus of points where condensation occurs is called the Wilson line.
- The Wilson line generally lies between 4 and 5 percent moisture curves in the saturation region



Flow of Steam through Nozzle

The phenomenon of super saturation



Flow of Steam through Nozzle

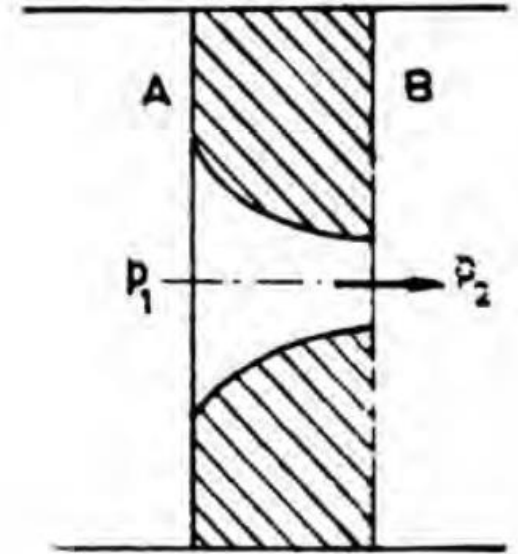
- **Effects of Super-saturation:**

The following are the effects of super-saturation in a nozzle.

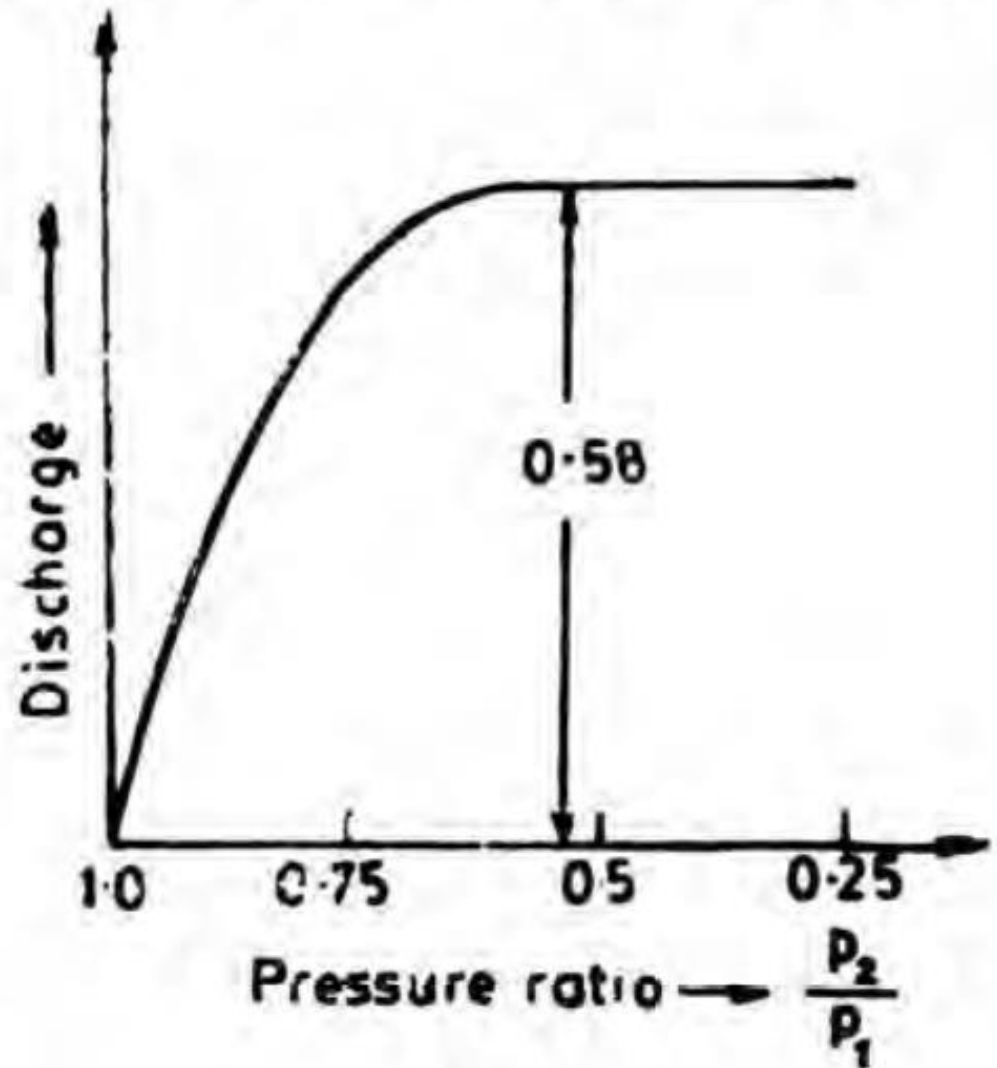
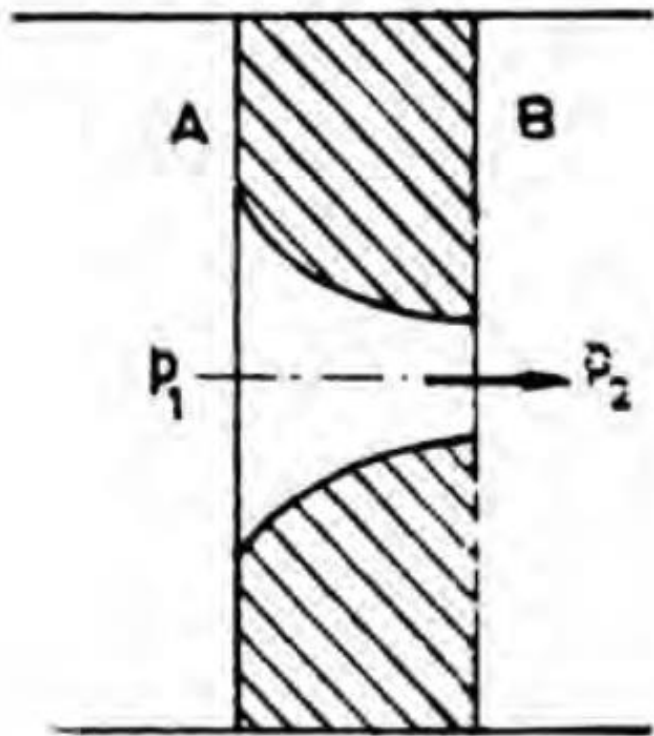
- The temperature at which the steam becomes supersaturated will be less than the saturation temperature corresponding to that pressure.
- Therefore, supersaturated steam will have the density more than that of equilibrium condition which results in the increase in the mass of steam discharged.
- Super-saturation causes the entropy of the steam to increase.
- Super-saturation reduces the heat drop. Thus the exit velocity of the steam is reduced.
- Super-saturation increases the dryness fraction of the steam.
- <https://www.youtube.com/watch?v=AOk0pEyvypw>

Effect of Back Pressure

- The flow in a nozzle is caused by a variation in pressure between two points (A and B)
- pressure at the exit (B) is referred to as the back-pressure
- pressure at the entry (A) is the stagnation pressure.
- ratio between them (P_2/P_1) is the back-pressure ratio, which can be used to control flow velocity.



Effect of Back Pressure

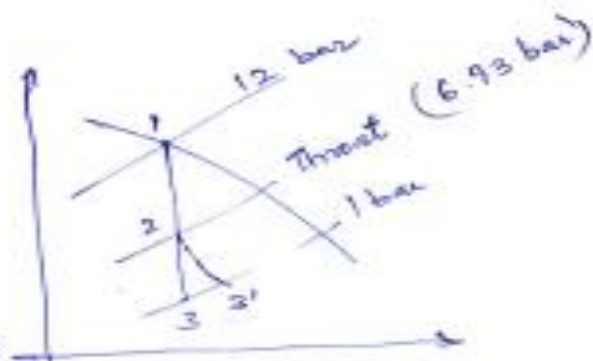


Dry saturated steam at a pressure of 12 bar flows through the nozzles at the rate of 4.5 kg/s and discharge at a pressure of 1 bar. The loss due to friction occurs only in the divergent portion of the nozzle and the magnitude is 10% of the total isentropic enthalpy drop. Determine the cross sectional area at the throat and exit of the nozzle.

Critical pressure ratio

$$\frac{P_2}{P_1} = \left(\frac{2}{n+1} \right)^{\frac{n}{n-1}}$$

where n isentropic expansion index



$n = 1.3$ for super heated steam
 $n = 1.135$ for Dry saturated steam
 $n = 1.035 + 0.1x$ for wet steam

Remember this

(Zeuner's eqn.)

$$\frac{P_2}{P_1} = \left(\frac{2}{1.135+1} \right)^{\frac{1.135}{0.135}} = 0.5774$$

$$\therefore P_2 = 0.5774 \times 12 = 6.93 \text{ bar}$$

$$h_1 = 2782.7 \text{ kJ/kg} \quad h_2 = ?$$

$$s_1 = 6.5194 = s_3$$

$$s_2 = 1.3027 + x(7.3598 - 1.3027)$$

$$x = 0.86$$

$$h_2 = h_f + x h_{fg} = 417.51 + 0.86(2675.4 - 417.51) = 2359.29 \text{ kJ/kg}$$

$$h_1 - h_3 = 423.41 \text{ kJ/kg}$$

$$h_1 - h_{3'} = 0.9 \times 423.41 = 381.07 \text{ kJ/kg}$$

$$h_{3'} = 2401.63 \text{ kJ/kg}$$

P kPa	T °C	v _f m³/kg	v _g m³/kg	v _{fg} m³/kg	h _f kJ/kg	h _g kJ/kg	h _{fg} kJ/kg	u _f kJ/kg	u _g kJ/kg	u _{fg} kJ/kg	s _f kJ/(kg K)	s _g kJ/(kg K)	s _{fg} kJ/(kg K)
100	99.63	0.001043	1.69	1.69	417.31	2675.27	2257.96	417.20	2505.90	2088.70	1.3020	7.3589	6.0569
125	105.99	0.001048	1.37	1.37	444.14	2685.08	2240.93	444.01	2513.25	2069.24	1.3734	7.2838	5.9104
150	111.37	0.001053	1.16	1.16	466.91	2693.20	2226.29	466.75	2519.34	2052.59	1.4330	7.2225	5.7895
175	116.06	0.001057	1.00	1.00	486.78	2700.13	2213.35	486.59	2524.54	2037.95	1.4843	7.1708	5.6865
200	120.23	0.001061	0.8855	0.8844	504.47	2706.17	2201.70	504.26	2529.07	2024.82	1.5295	7.1260	5.5966
225	123.99	0.001064	0.7930	0.7920	520.46	2711.52	2191.06	520.22	2533.09	2012.86	1.5699	7.0866	5.5167
250	127.43	0.001067	0.7185	0.7175	535.09	2716.32	2181.23	534.83	2536.69	2001.86	1.6065	7.0513	5.4448
275	130.60	0.001070	0.6571	0.6561	548.60	2720.67	2172.07	548.30	2539.95	1991.65	1.6400	7.0194	5.3794
300	133.54	0.001073	0.6057	0.6046	561.16	2724.63	2163.47	560.84	2542.93	1982.09	1.6710	6.9903	5.3193
325	136.29	0.001076	0.5619	0.5608	572.92	2728.28	2155.36	572.57	2545.67	1973.10	1.6997	6.9635	5.2637
350	138.87	0.001079	0.5241	0.5230	583.98	2731.64	2147.66	583.61	2548.20	1964.59	1.7266	6.9386	5.2120
375	141.31	0.001081	0.4913	0.4902	594.44	2734.77	2140.33	594.04	2550.55	1956.51	1.7518	6.9155	5.1637
400	143.62	0.001084	0.4624	0.4613	604.37	2737.68	2133.31	603.93	2552.74	1948.80	1.7756	6.8938	5.1182
425	145.81	0.001086	0.4368	0.4357	613.82	2740.41	2126.59	613.36	2554.79	1941.43	1.7982	6.8735	5.0753
450	147.91	0.001088	0.4139	0.4128	622.85	2742.97	2120.12	622.36	2556.71	1934.36	1.8196	6.8543	5.0347
475	149.91	0.001091	0.3934	0.3923	631.49	2745.37	2113.88	630.97	2558.52	1927.55	1.8400	6.8361	4.9961
500	151.84	0.001093	0.3748	0.3737	639.79	2747.64	2107.86	639.24	2560.23	1920.99	1.8595	6.8188	4.9594
525	153.68	0.001095	0.3580	0.3569	647.77	2749.79	2102.02	647.19	2561.85	1914.65	1.8781	6.8024	4.9243
550	155.46	0.001097	0.3426	0.3415	655.46	2751.83	2096.37	654.86	2563.38	1908.52	1.8960	6.7867	4.8907
575	157.17	0.001099	0.3286	0.3275	662.89	2753.76	2090.87	662.26	2564.83	1902.58	1.9132	6.7717	4.8585
600	158.83	0.001101	0.3156	0.3145	670.07	2755.60	2085.52	669.41	2566.22	1896.80	1.9298	6.7573	4.8275
625	160.43	0.001103	0.3037	0.3026	677.03	2757.35	2080.32	676.34	2567.54	1891.19	1.9458	6.7435	4.7977
650	161.98	0.001105	0.2927	0.2915	683.78	2759.02	2075.24	683.06	2568.79	1885.73	1.9612	6.7302	4.7690
675	163.49	0.001106	0.2824	0.2813	690.33	2760.61	2070.28	689.58	2569.99	1880.41	1.9762	6.7174	4.7412
700	164.95	0.001108	0.2728	0.2717	696.70	2762.13	2065.43	695.92	2571.14	1875.22	1.9906	6.7050	4.7144
725	166.37	0.001110	0.2639	0.2628	702.90	2763.59	2060.69	702.09	2572.24	1870.15	2.0047	6.6931	4.6884
750	167.75	0.001112	0.2556	0.2545	708.93	2764.99	2056.05	708.10	2573.29	1865.19	2.0183	6.6815	4.6632
775	169.09	0.001113	0.2478	0.2467	714.82	2766.33	2051.51	713.96	2574.30	1860.35	2.0315	6.6703	4.6388
800	170.40	0.001115	0.2404	0.2393	720.57	2767.61	2047.05	719.67	2575.27	1855.60	2.0444	6.6595	4.6151

P kPa	T °C	v _f m³/kg	v _g m³/kg	v _{fg} m³/kg	h _f kJ/kg	h _g kJ/kg	h _{fg} kJ/kg	u _f kJ/kg	u _g kJ/kg	u _{fg} kJ/kg	s _f kJ/(kg K)	s _g kJ/(kg K)	s _{fg} kJ/(kg K)
825	171.68	0.001117	0.2335	0.2324	726.18	2768.85	2042.67	725.25	2576.20	1850.95	2.0570	6.6490	4.5920
850	172.93	0.001118	0.2270	0.2259	731.66	2770.03	2038.37	730.71	2577.10	1846.39	2.0692	6.6388	4.5696
875	174.15	0.001120	0.2208	0.2197	737.02	2771.17	2034.15	736.04	2577.96	1841.92	2.0811	6.6288	4.5477
900	175.35	0.001121	0.2150	0.2139	742.27	2772.27	2030.00	741.26	2578.79	1837.53	2.0927	6.6191	4.5264
925	176.51	0.001123	0.2094	0.2083	747.41	2773.33	2025.92	746.37	2579.59	1833.22	2.1041	6.6097	4.5056
950	177.66	0.001124	0.2042	0.2031	752.45	2774.35	2021.90	751.38	2580.36	1828.98	2.1152	6.6005	4.4853
975	178.78	0.001126	0.1992	0.1981	757.39	2775.33	2017.95	756.29	2581.10	1824.82	2.1261	6.5916	4.4655
1000	179.87	0.001127	0.1945	0.1933	762.23	2776.28	2014.05	761.10	2581.82	1820.72	2.1367	6.5828	4.4462
1250	189.80	0.001141	0.1571	0.1560	806.32	2784.16	1977.84	804.89	2587.79	1782.90	2.2321	6.5052	4.2730
1500	198.28	0.001154	0.1318	0.1307	844.32	2789.84	1945.52	842.59	2592.12	1749.53	2.3127	6.4407	4.1280
1750	205.72	0.001165	0.1135	0.1124	877.95	2793.99	1916.04	875.91	2595.31	1719.40	2.3828	6.3854	4.0026
2000	212.37	0.001176	0.099676	0.098500	908.29	2797.03	1888.74	905.94	2597.67	1691.74	2.4450	6.3367	3.8917
2250	218.40	0.001187	0.088799	0.087612	936.04	2799.20	1863.16	933.37	2599.41	1666.04	2.5012	6.2931	3.7920
2500	223.94	0.001197	0.080025	0.078828	961.70	2800.70	1839.01	958.71	2600.64	1641.94	2.5524	6.2535	3.7010
2750	229.06	0.001206	0.072792	0.071586	985.63	2801.67	1816.03	982.31	2601.49	1619.17	2.5997	6.2170	3.6173
3000	233.84	0.001216	0.066724	0.065508	1008.11	2802.18	1794.07	1004.47	2602.01	1597.55	2.6437	6.1832	3.5396
3250	238.32	0.001225	0.061557	0.060332	1029.35	2802.33	1772.98	1025.37	2602.27	1576.90	2.6848	6.1517	3.4669
3500	242.54	0.001234	0.057103	0.055869	1049.52	2802.17	1752.65	1045.20	2602.31	1557.11	2.7234	6.1220	3.3985
3750	246.54	0.001243	0.053222	0.051979	1068.75	2801.74	1732.99	1064.09	2602.16	1538.07	2.7600	6.0939	3.3339
4000	250.34	0.001252	0.049808	0.048557	1087.16	2801.09	1713.93	1082.15	2601.86	1519.71	2.7947	6.0672	3.2725
4250	253.96	0.001260	0.046782	0.045522	1104.83	2800.24	1695.41	1099.48	2601.41	1501.94	2.8277	6.0417	3.2140
4500	257.42	0.001269	0.044080	0.042811	1121.85	2799.22	1677.37	1116.14	2600.86	1484.71	2.8593	6.0174	3.1581
4750	260.73	0.001277	0.041652	0.040375	1138.28	2798.04	1659.76	1132.21	2600.19	1467.98	2.8895	5.9940	3.1044
5000	263.92	0.001286	0.039458	0.038173	1154.18	2796.74	1642.55	1147.75	2599.44	1451.69	2.9186	5.9714	3.0528
5250	266.98	0.001294	0.037466	0.036171	1169.60	2795.31	1625.71	1162.81	2598.62	1435.81	2.9466	5.9497	3.0031
5500	269.94	0.001303	0.035648	0.034345	1184.58	2793.78	1609.20	1177.41	2597.72	1420.30	2.9736	5.9286	2.9550
5750	272.80	0.001311	0.033982	0.032671	1199.16	2792.16	1593.00	1191.62	2596.76	1405.14	2.9998	5.9082	2.9084
6000	275.56	0.001319	0.032449	0.031130	1213.37	2790.45	1577.08	1205.45	2595.75	1390.30	3.0251	5.8883	2.8633
6250	278.23	0.001328	0.031035	0.029708	1227.24	2788.66	1561.42	1218.94	2594.69	1375.75	3.0496	5.8690	2.8194

$$h_2 =$$

$$\alpha_2$$

$$S_2 = S_f + \alpha \cdot S_{fg} @ 6.93 \text{ bar}$$

$$\alpha_2 = 0.9606$$

$$h_2 = 677.06 + 0.9606 \times 2064.9, \quad b_2 = 0.0011082 + 0.9606 \times (0.27268 - 0.001082) \\ = 2680.60 \text{ kJ/kg} \quad \cdot 0.26198 \text{ m}^3/\text{kg}$$

$$V_2 = \sqrt{2(h_1 - h_2)} = \sqrt{2(2782 - 2680.6) \times 1000} = 450.33 \text{ m/s}$$

$$V_3' = \sqrt{2(h_1 - h_3')} = \sqrt{2 \times 381.07 \times 1000} = 873.01 \text{ m/s}$$

$$m = \frac{A_2 V_2}{v_2} \quad A_2 = \frac{m v_2}{V_2} = \frac{4.5 \times 0.26198}{450.33} = 0.002618 \text{ m}^2 \\ = 26.18 \text{ cm}^2$$

$$A_3' = \frac{m v_3'}{V_3'}$$

$$\alpha_3' = 0.8787$$

$$b_3' = 0.0010434 + 0.8787 \times 1.6737 \\ = 1.489 \text{ m}^3/\text{kg}$$

$$A_3 = \frac{4.5 \times 1.489}{873.01} = 0.007675 \text{ m}^2 = 76.75 \text{ cm}^2$$

Steam at a pressure of 10 bar and 95% dry is supplied through a CD nozzle to the wheel chamber, where the pressure is maintained at 0.12 bar. The mass flow rate through the nozzle is 8.16 kg/kWh and the work developed by the wheel is 110 kW.

Determine

- a) pressure at the throat using Zeuners equation
- b) No. of nozzles needed if each nozzle has a throat diameter of 5 mm
- c) c) the exit diameter of the nozzle if 10% of the overall enthalpy drop overcomes the friction in the divergent portion

By Zeuner's eqn., $n = 1.035 + 0.1 x$

$$n = 1.035 + 0.1 \times 0.95 = 1.13$$

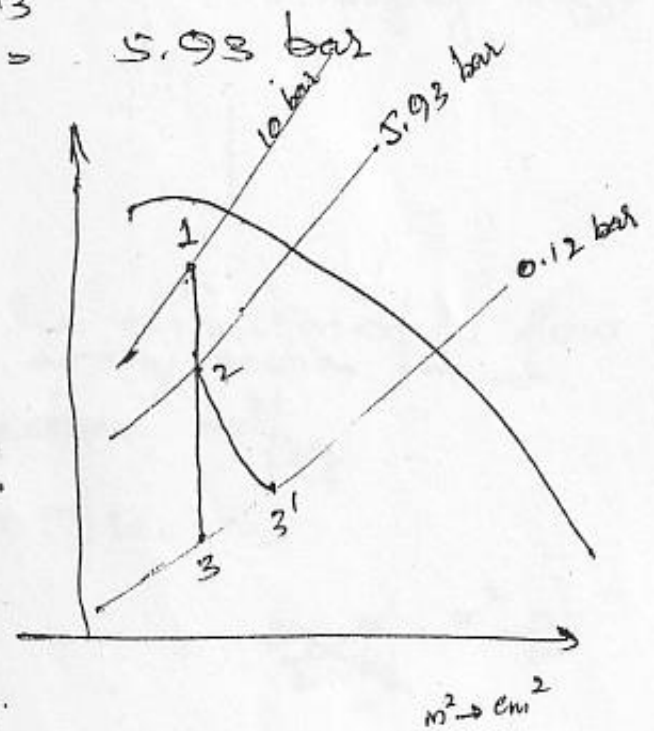
$$P_2 = P_1 \cdot \left(\frac{2}{n+1} \right)^{\frac{n}{n-1}} = 10 \times \left(\frac{2}{1.13+1} \right)^{\frac{1.13}{0.13}} = 5.93 \text{ bar}$$

$$\eta_{\text{nozzle}} = 90\% \quad \text{or} \quad 0.9 = \frac{h_1 - h_3}{h_1 - h_2}$$

$$m = \frac{8.16 \times 110}{3600} = 0.25 \text{ kg/sec}$$

$$V_2 = \sqrt{2 \times 1000 (h_1 - h_2)}$$

$$= \sqrt{2 \times 1000 \times (2675 - 2580)} = 436 \text{ m/sec.}$$



Combined area of throat $A_2 = \frac{m V_2}{V_2} = \frac{0.25 \times 0.305 \times 10^4}{436} = 1.74 \text{ cm}^2$

Throat area per nozzle $= \frac{\pi}{4} (0.5)^2 \text{ cm}^2 = 0.1965 \text{ cm}^2$

No. of nozzles $= \frac{1.74}{0.1965} = 8.85 = \underline{\underline{9}}$

$$\text{Nozzle exit velocity } V_3' = \sqrt{2 \times 1000 (h_1 - h_3')} = \sqrt{2 \times 1000 \times 575} = 1072 \text{ m/s.}$$

$$\text{Combined area of exit} = \frac{m V_3'}{V_3'} = \frac{0.25 \times 10 \times 10^4}{1072} = 23.32 \text{ cm}^2$$

$$\text{Area of 1 nozzle} = \frac{23.32}{9} = 2.59 \text{ cm}^2$$

$$\therefore \text{Nozzle dia at exit} = \sqrt{\frac{2.59 \times 4}{\pi}} = \underline{\underline{1.81 \text{ cm}}}$$

